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Semester: 7

Lab Assignment: Introduction to Object-Oriented PHP

Lab Topic

Practicing Classes, Objects, Constructors, Access Modifiers, and Inheritance

Part A — Simple Class and Object

```
<?php
class Student {
 function sayHello() {
echo ‘Hello I am a student.’:
}
}
                    // Create an object here
$student1 = new student():
                    // Call the method here
$student1 ->sayHello():
?>
```

*** According to this code, I have created a class, object, method and called the method.

Part B — Class with Constructor

```
<?php
class Student {
public $name;
 public $studentId;
 public $department;

function __construct($name, $studentId, $department) {
 // Store the values in the properties
$this ->name = $name;
$this ->studentid = $studentid;
$this ->department = $department;
 }
function showInfo() {
 // Display student information
Echo ‘name; ’. $this ->name .
‘<br>’;
Echo ‘student id; ’. $this ->student id .
‘<br>’;
Echo ‘department; ’. $this department;
}
}
```

```
$student1 = new Student( "Ahmad", 1001, "Computer Science" );
$student1->showInfo();
?>
```

Expected Output

Name: Ahmad Student ID: 1001 Department: Computer Science

Part C — Create Another Object

Using the same Student class, create another student.

```
$student2 = new student('Sara', 1002, 'information system');
$student2 ->showInfo();
```

Expected Output

Name: Sara Student ID: 1002 Department: Information Systems

How many classes did you create?

Answer: 1 class, I created 1 class called student and multiple object from it.

How many objects did you create?

Answer: 2 objects I have created, called $student1, $student2

Part D — Access Modifiers

```
<?php
class BankAccount {
 public $ownerName;
private $balance;
function __construct($ownerName, $balance) {
```

```
 // Store the values
$this ->ownerName = $ownerName;
$this ->balance = $balance;
 }
function showBalance() {
// Display the balance
Echo ‘balance; ’. $this ->balance;
}
}
$account1 = new BankAccount( "Ahmad", 5000 );
echo "Owner: " . $account1->ownerName .
 "<br>";
$account1->showBalance();
?>
```

Expected Output

Owner: Ahmad Balance: 5000

Expected Output

Owner: Ahmad

Balance: 5000

Try this;

After running the program successfully, try:

```
echo $account1->balance;
```

Question

Does it work?

Yes / No

Answer: No

Why?

Because balance is defined private , we cannot access to that from outside the class.

Part E — Simple Inheritance

```
<?php
class Person {
public $name;
function __construct($name) {
 // Store the name

}
function introduce() {
 // Display the person's name
$this -> name = $name;
}
Function introduce(){
Echo ‘my name is ‘. $this –>name.
‘<br>’;
}
```

```
}
class Student extends Person { function study() {
// Display that the student is studying
Echo $this ->name . ‘is studying .’;
 }
}
$student1 = new Student("Ahmad");
$student1->introduce();
$student1->study();
?>
```

Expected Output

My name is Ahmad

Ahmad is studying.

Part F — Understanding Inheritance

Answer the following questions.

Question 1

Which class is the parent class?

Answer:

Person is the parent class.

Question 2

Which class is the child class?

Answer:

Student is the child class.

Question 3

Which keyword creates the inheritance relationship?

Answer:

Extends is used in php for inheritance.

Question 4

The introduce() method was written inside which class?

Answer:

It is written inside the person class.

Question 5

Can the $student1 object call introduce() ?

Answer:

Yes, because student inherits from person and can use all public method of parent.

Question 6

Why can $student1 use introduce() even though it is not written inside the Student class?

Answer:

because student inherits from person and can use all public method of parent.

Part G — Small Independent Exercise

Create your own simple OOP program using Vehicle and Car.

```php
<?php

class Vehicle
{
    protected $brand;

    function __construct($brand)
    {
        $this->brand = $brand;
    }

    function start()
    {
        echo "The vehicle is starting.<br>";
    }
}

class Car extends Vehicle
{
    function showBrand()
    {
        echo "Car brand: " . $this->brand;
    }
}

$car1 = new Car("Toyota");

$car1->start();

$car1->showBrand();
```

?>

Output

The vehicle is starting

Car brand: Toyota